

# INTEGRATION AS INVERSE PROCESS OF DIFFERENTIATION

## Additional HOMEWORK

1.  $\int (\tan x + \cot x)^2 dx$
2.  $\int \cos(\pi\theta + 1) d\theta$
3.  $\int \frac{\cos 5x + \cos 4x}{1 - 2 \cos 3x} dx$
4.  $\int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx$
5.  $\int \frac{\cos 2x - \cos 2\alpha}{\sin x - \sin \alpha} dx$
6.  $\int \frac{\cos x - \cos 2x}{1 - \cos x} dx$
7.  $\int \frac{\sec 2x - 1}{\sec 2x + 1} dx$
8.  $\int \sqrt{1 - \sin x} dx$
9.  $\int \frac{\cos x - \sin x}{\cos x + \sin x} (2 + 2 \sin 2x) dx$
10.  $\int \sqrt{1 - \cos x} dx$
11.  $\int (3 \sin x \cos^2 x - \sin^3 x) dx$
12.  $\int \frac{dx}{1 + \cos x}$
13.  $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x} dx$
14.  $\int \frac{\cos^4 x - \sin^4 x}{\sqrt{1 + \cos 4x}} dx$
15.  $\int \frac{\cos x}{\sin^2 x} (1 - 3 \cos^3 x) dx$
16.  $\int \sin mx \sin nx dx$
17.  $\int \sin^{-1}(\cos x) dx$
18.  $\int \tan^{-1} \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} dx$
19.  $\int \tan^{-1}(\sec x + \tan x) dx$
20.  $\int \tan^{-1} \sqrt{\frac{1 - \sin x}{1 + \sin x}} dx$
21.  $\int \tan^{-1} \left( \frac{\sin 2x}{1 + \cos 2x} \right) dx$
22.  $\int \cos^{-1}(\sin x) dx$
23.  $\int \cot^{-1} \left( \frac{\sin 2x}{1 - \cos 2x} \right) dx$
24.  $\int \sin^{-1} \left( \frac{2 \tan x}{1 + \tan^2 x} \right) dx$
25.  $\int (a \tan x + b \cot x)^2 dx$
26.  $\int \frac{5 \cos^3 x + 6 \sin^3 x}{2 \sin^2 x \cos^2 x} dx$
27.  $\int \left( 3 \sin x - 4 \cos x + \frac{5}{\cos^2 x} - \frac{6}{\sin^2 x} + \tan^2 x - \cot^2 x \right) dx$
28.  $\int \left( \frac{5 \cos^3 x + 2 \sin^3 x}{2 \sin^2 x \cos^2 x} + \sqrt{1 + \sin 2x} + \frac{1 + 2 \sin x}{\cos^2 x} + \frac{1 - \cos 2x}{1 + \cos 2x} \right) dx$
29.  $\int \left( \frac{1 - x^4}{1 - x} - \frac{6}{x \sqrt{x^2 - 4}} + \tan x + \cot x \right) dx$
30.  $\int \left( \tan^{-1} \left( \frac{\log \left( \frac{e}{x^4} \right)}{\log (ex^2)} \right) - \tan^{-1} \left( \frac{\log (ex^6)}{\log \left( \frac{e^3}{x^2} \right)} \right) \right) dx$